Rohan Paramane
9+ years

CJ
OOP
Language Fundametals
Exception Handling
Collections
File IO
Function Programming
Java 8 Interfaces
Reflection , Annotation
MultiThreading

OOP ?
Abstraction, Encapsulation,Inheritance,Polymorphism

OOP
Major Pillars
- Abstraction
- Encapsulation
- Modularity
- Hirerachy

Minor Pillars
- Typing/Polymorphism
- Concurrency
- Persistance

For any OOP, it is compulsary to follow all the major pillars

#Major Pillars

1. Abstraction
    Getting to know only the essential deatils
    - Examples
        Calling a function
        Creating the objects

printf("%d",n1)

2. Encapsulation
    Binding the things toether
    Binding of data and code together
    - Examples
        Defining the functions
        Defining the classes/struct

Manafacturer
Encapsulation

Consumer
Abstraction

3. Modularity
    - Dividing a task into smaller tasks
    - Examples
        Dividing the program into multiple files
        Defining multiple classes and multiple functions
        are also example of modularity

```
demo.c
#include<stdio.h>
int main(){
printf("Enter 2 nos")
scanf(%d%d)
printf("Addition")
}
```

add
sub
mul
div

4. Hirerachy (Imp topic of oop)
    - Reusability
    - It can be implemented by using
        - Association (has-a relationship)
        - Inheritance (is-a relationship)

feature

SIM
Network
Anntena

smartphone
Android OS -> google
Customize

```
class Date{
        int day;
        int month;
        int year

        // Implemented
}
```

```
class Person{
String name;
Date dob; // Association

// Implementation

}
```

```
// Inheritance
class Employee extends Person{
double salary
}
```

Person Date	Person has-a Date	Person is-a Date
Employee Person	Employee has-a Person	Employee is-a Person

Minor Pillars

- These Pillars are optional to be implemented in the oop languages

1. Polymorphism
    - One entity that takes multiple forms
    - Exmples - Mobile, Person
    - Example
        Function/Method overloading (Compile Time Polymorphism)
        Function/Method overriding (RunTime Polymorphism)

2. Concurrency
    - Executing the multiple tasks at the same time
    - Multiple Threads (MultiThreading)

3. Persistance
    - Storing the data permanantly
    - File IO / JDBC

Installation

- JDK -> Java Development Kit
    - To download click on below link
        https://adoptium.net/en-GB/temurin/releases?version=11&os=any&arch=any
    - Make sure to downdload the JDK 11 and install it

- Editor -> Limitations
    - Notepad, vscode,....etc

- IDE -> Integrated Development Environment
    - STS, intellij , eclipse

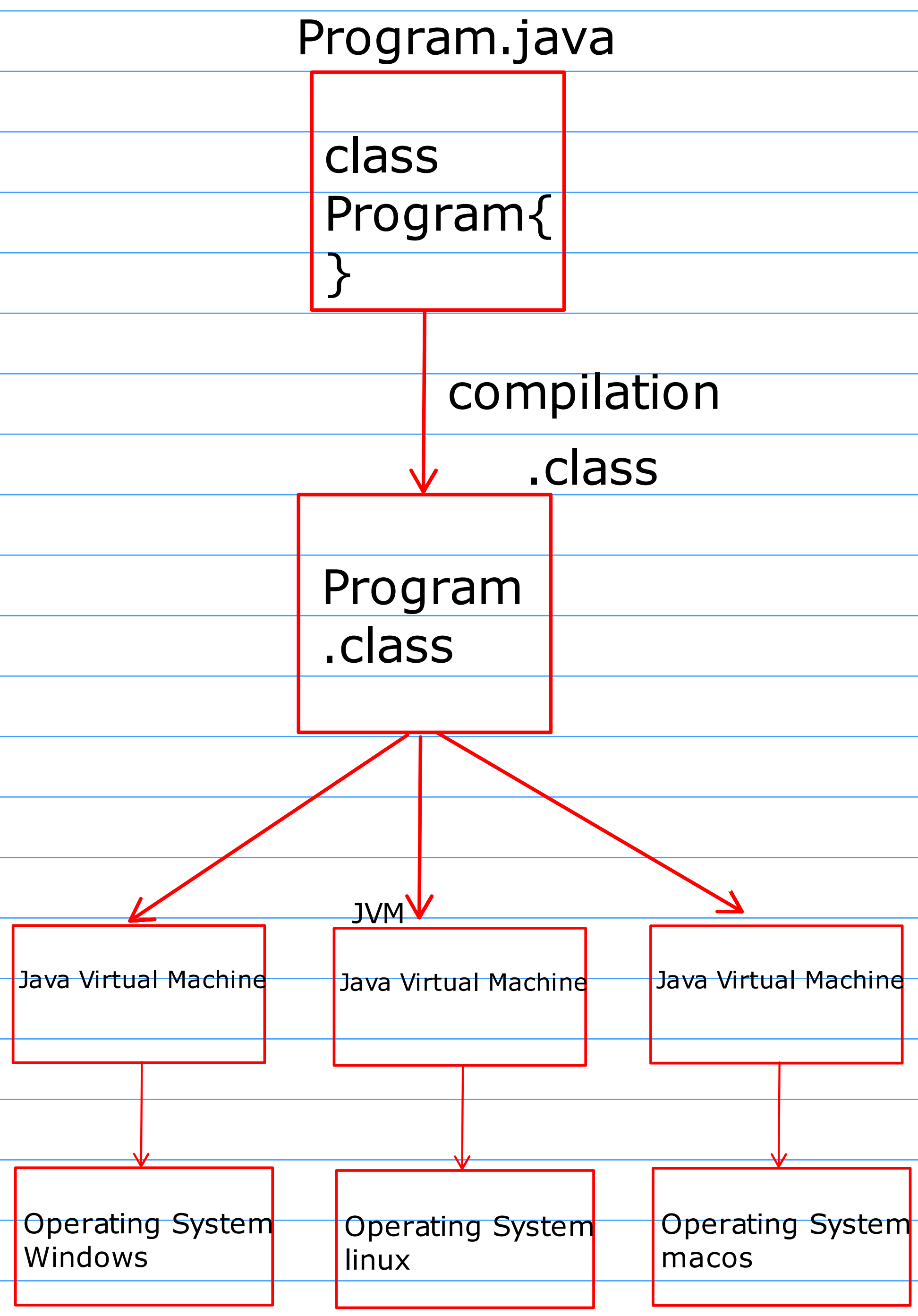
JDK -> Java Developmet Kit (jdk 11)
JDK = tools + docs + JRE (Java RunTime Environment) JRE
JRE = rt.jar(core libraries) + JVM (Java Virtual Machine)

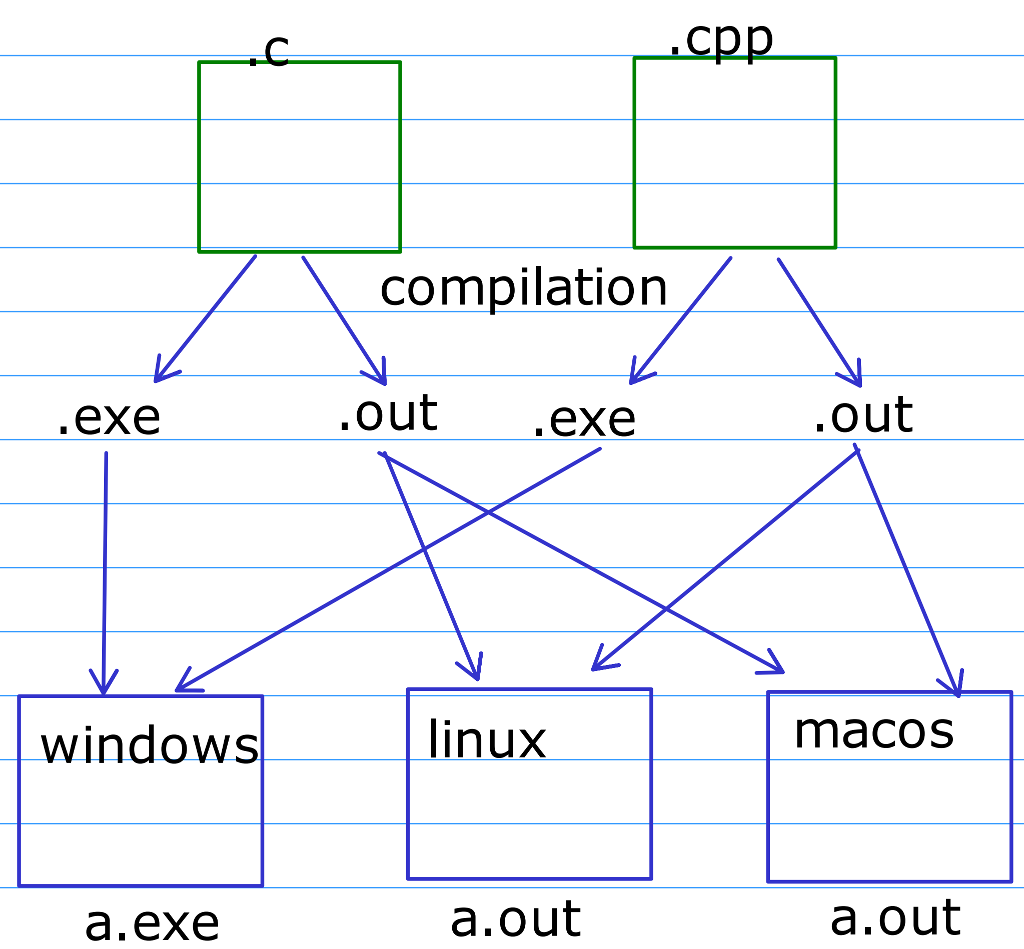
System.out.println
System -> Class
out -> It is an object of an PrintStream class
pritntln -> It is the method of the PrintStream class

- When the java code is compiled it creates a .class file
- .class file contains the bytecode
- bytecode is an intermediate code generated by the java compiler that is understandable only by the JVM

javac -> java compiler
java -> executer

javac Program01.java -> Program01.class
java Program01 -> java <name of .class file>

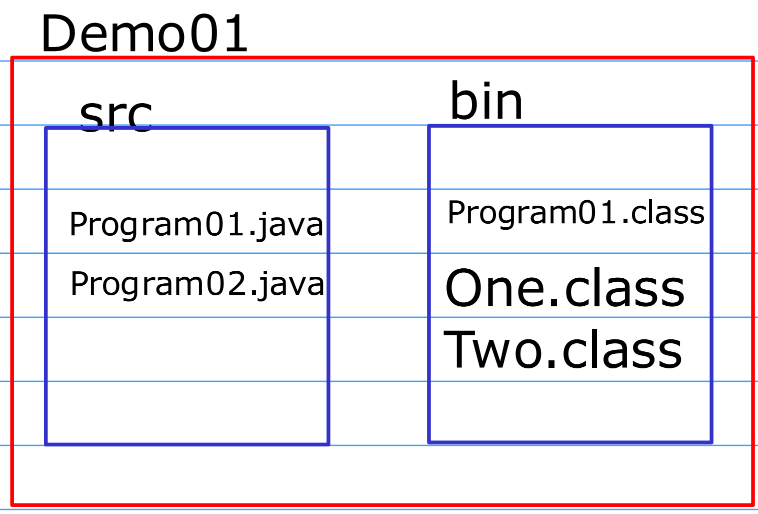




WORA
Write Once Run Anywhere

- When we compile tha java code it will create as many no of classes as defined inside that java file

- Project Structure in java



- open terminal from the src directory
- for compilation and storing the .class file in bin use below cmd
- javac -d ..\bin Program01.java
- set the classpath to tell the jvm where .class files are stored
- SET CLASSPATH=..\bin
- execute the code
- java Program01

Homework -> Create the Program02 example and test it

PATH
- It is an opeating System variable

CLASSPATH
- It is an JVM varaiable
- It is used to store the path of .class files